

# NEUROSCIENCE MIDTERM

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## MIDTERMS LESSON 1: VENTRICLES, COVERINGS, AND VASCULAR SUPPLY OF THE BRAIN

### VENTRICULAR SYSTEM

- Communicating system of cavities that are **lined with Ependyma** and **filled with Cerebrospinal Fluid** (to provide nutrients and protection).

**LATERAL VENTRICLE** - largest of the ventricles.

- **2 central portions** (body and atrium)
- **3 extension** (horns)
- **CHOROID PLEXUS** - **produces csf**; fringe-like vascular process of pia mater containing **capillaries of the choroid arteries**.
- **TELA CHOROIDEA** - attachment of the plexus to adjacent brain structures.
- **INTERVENTRICULAR FORAMEN (FORAMEN OF MONRO)** - communication between **lateral and third ventricle**.

**THIRD VENTRICLE** - narrow vertical cleft between the 2 halves of diencephalon.

- **OPTIC RECESS** - extension of 3rd ventricle **between lamina terminalis and optic chiasm**.
- **PINEAL RECESS** - projects into the **stalk of the pineal body**; secretes **melatonin**.

**CEREBRAL AQUEDACT (AQUEDACT SYLVIIUS)** - narrow curved channel running from posterior 3rd ventricle into 4th; **no csf existing (choroid plexus)**.

**FOURTH VENTRICLE** - pyramid shaped cavity **bounded ventrally by pons and medulla oblongata**.

- **MEDIAL APERTURE (FORAMEN OF MAGENDIE)** - opening of caudal portion of roof of the ventricle.
- **LATERAL APERTURE (FORAMEN OF LUSCHKA)** - opening of lateral recess into the subarachnoid space near the flocculus of the cerebellum.

### MENINGES AND SPACES

**SUBDURAL SPACE** - separates the **Dura mater superiorly** and **arachnoid mater inferiorly**.

- **Contains a few drops of CSF.**

**SUBARACHNOID SPACE** - **inferior to the arachnoid mater** and **superior to the pia mater**.

- Contains CSF and major arteries of the brain
- **Types of stroke: Ischemic and Hemorrhagic stroke.**

**LEPTOMENINGES** - composed by **pia and dura mater**.

**PERIVASCULAR SPACE (VIRCHOW-ROBIN'S SPACE)** - narrow extension of subarachnoid space along the pia mater.

- Vessels deep into the brain tissue.

**DURA MATER (PACHYMENINX)** - tough fibrous structure with **inner (meningeal)** and **outer (periosteal) layer**.

- Most dura's venous sinuses lie between dura layers:

- **OUTER PERIOSTEAL LAYER (skull)** - firmly attached to the inner surface of the cranial bones and sends vascular and fibrous extensions into the bone itself.
- **INNER MENINGEAL LAYER** - continuous with the spinal dura.

### TENTORIUM CEREBELLI

- **Separates occipital lobes from cerebellum**; roughly transverse membrane.
- Fuses with Falx Cerebri towards midline.
- Attachments:
  - Rear and side of the skull at the **transverse sinuses**
  - At front - **petrous portion of the temporal bone** and to the **clinoid process of the sphenoid bone**.
- **TENTORIAL NOTCH (INCISURA TENTORIA)** - free, curved anterior border leaves a large opening for **passage of the upper brain stem, aqueduct, and vessels**.

### FALX CEREBELLI

- **Separates 2 cerebral hemispheres**; projects between cerebellar hemispheres from the inner surface of occipital bone to form a small triangular dural septum.
- Attached to the **inner surface of the skull midplain**, from the **crista galli to the internal occipital protuberance** (becomes continuous with the **tentorium cerebelli**).
- **Dura septa** - extends down into the longitudinal fissure between the cerebral hemispheres.

### DIAPHRAGMA SELLAE

- Forms an incomplete lid over the hypophysis in the sella turcica by connecting the clinoid attachments of the 2 sides of the tentorium cerebelli.
- **Where pituitary stalk passes through.**

### ARACHNOID MATER

- Delicate, avascular membrane that covers the **subarachnoid space**.
- Inner surface of arachnoid is connected to pia by fine **ARACHNOID TRABECULAE**
- Closely covers the inner surface of dura mater but is separated from it by **subdural space** (contains a thin film of fluid).

**ARACHNOID GRANULATIONS** - sites of **absorption of the CSF**.

- Consists of many **microscopic villi**
- **Berry-like clumps** protruding into the superior sagittal sinus (vein).

**SUBARACHNOID SPACE** - between arachnoid and pia mater; narrow in cerebral hemispheres that becomes wider in areas at the base of the brain.

- Once it's widened, it's called **CISTERN**.
- **CISTERNA MAGNA** - from bridging of the arachnoid over space between medulla and cerebral hemispheres; continuous with **spinal subarachnoid space**.
- **PONTINE CISTERN** - ventral aspect of the pons that contains the basilar artery and some veins.

**WIDE SPACES BELOW CEREBRUM THAT SEPARATES TEMPORAL LOBE:**

1. **CHIASMATIC CISTERNA** - above optic chiasm
2. **SUPRASELLAR CISTERNA** - above diaphragmatic sellae
3. **INTERPENDICULAR CISTERN** - between cerebral peduncles

PIA MATER

- Thin connective tissue membrane that covers the brain that **extends to sulci and fissures**.
- Extends into the transverse cerebral fissure under the Corpus Callosum
- **Forms the Tela Choroidea of the 3rd and lateral ventricles** and combines with the ependyma + choroid vessels to form **the choroid plexus of these ventricles**.

CLINICAL CORRELATION

**THE TENTORIUM** - dural septum that separates the **supratentorial and infratentorial compartments** (communicates through incisura (contains midbrain)).

- mass/lesion can force brain tissue through the incisura.

**TRANSTERTORIAL HERNIATION (UNCAL)** - the uncus (medial temporal lobe) is pushed through the tentorium, compressing the brainstem and CN III (Oculomotor)

- **Clinical sign: Ipsilateral dilated pupil and third nerve paresis.**

**SUBFALCINE HERNIATION** - cingulate gyrus is displaced into or under the falx cerebri.

- **Clinical sign: Lateral displacement of midline structures.**

**CONING (TONSILAR)** - cerebellar tonsils are forced downward into the foramen magnum.

- **Clinical sign: Rapid brainstem compression; frequently fatal.**

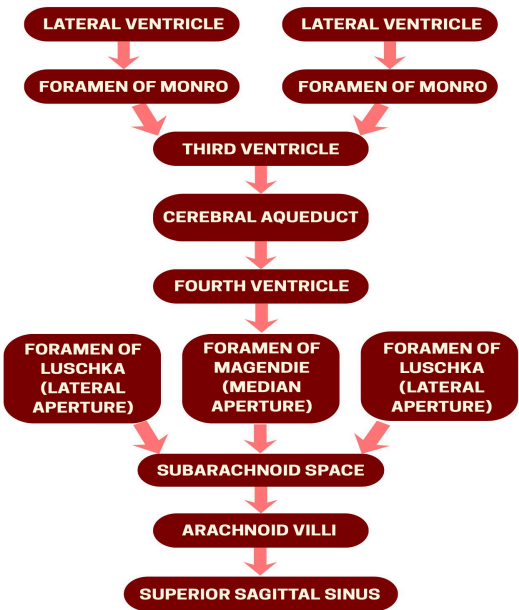
CEREBROSPINAL FLUID

- Mechanical support of the brain and **acts like a protective water jacket**.
- Controls brain excitability by **regulating ionic compositions**; carries away metabolites.
- Clear and colorless.

CSF PRESSURE

- **Normal mean - 70-180mmH2O**
- **Increase**
  - Intracranial volume (**mass/tumors or with massive infarcts that can cause swelling**)
  - blood volume (**hemorrhage**)
  - CSF volume (**Hydrocephalus**)

CSF CIRCULATION:



CSF CLINICAL CORRELATION

- Blocking the circulatory pathway of CSF leads to **hydrocephalus**.

2 types of hydrocephalus:

1. **NON-COMMUNICATING (OBSTRUCTIVE)** - most common
  - Ventricles **cannot reach subarachnoid space**
  - Block rapidly leads to **dilatation of one or more ventricles**.
  - **gyri may flatten against the skull**; can enlarge if the skull is pliable.
- **CEREBRAL AQUEDUCT** - most common site of obstruction
2. **COMMUNICATING HYDROCEPHALUS** - obstruction in **arachnoid space**
  - Can be a result of prior bleeding/meningitis
  - If intracranial pressure is raised because of CSF, the central canal of the spinal cord may dilate.
  - Spaces filled by CSF are **uniformly enlarged** without increase in intracranial pressure.
- **NORMAL PRESSURE HYDROCEPHALUS** - may be accompanied by a **gait disorder, incontinence, and dementia** (in elderly).

BARRIERS IN THE NERVOUS SYSTEM

BLOOD BRAIN BARRIER

- **Formed by:**

- **BLOOD-CSF BARRIER** - 60% is formed by active transport from blood vessels in the choroid plexus.
  - **Epithelial cells of plexus** - forms a continuous layer that **selectively permits** the passage of some substance (joined by tight junctions).
- **VASCULAR-ENDOTHELIAL BARRIER** - for promotion of exchange of oxygen, carbon dioxide, amino acids, and sugars between blood and brain.
  - Maintained by tight junctions in between.
- **ARACHNOID BARRIER** - substances diffusing out of dural vessels do not enter the CSF of subarachnoidal space.

EPENDYMA - produces and circulates CSF.

- Continuous with the epithelium of the choroid plexus.

SKULL

- Rigid in adults; pliable in infants
- Surrounds the brain and meninges completely and forms a strong mechanical protection.
  - *In adults, the volume of the brain can increase beyond the capacity of the intact cranium as a result of swelling after injury.*

SKULL (BASAL VIEW)

- **ANTERIOR PORTION OF THE BASE OF THE SKULL**
  - Hard palate
    - Below level of remainder of the inferior skull surface
- **POST PORTION OF SKULL** - Choanae
- **LATERAL TO CHOANAE** - Pterygoid plates
- **FORAMEN OVALE** - base of the pterygoid plates.
  - 3rd branch of trigeminal nerve
  - Accessory meningeal artery
  - Superficial petrosal nerve
- **FORAMEN SPINOSUM** - middle meningeal vessels; posterior to foramen ovale.
- **STYLOMASTOID FORAMEN** - base of styloid process; facial nerve exits.
- **FORAMEN LACERUM** - large irregular aperture at the base of the medial pterygoid plate.
  - Where the Internal Carotid Artery (ICA) crosses.
- **SULCUS TUBAE AUDITIVE** - contains cartilaginous part of auditory tube.
  - Forms **bony part of the auditory tube**.
- **JUGULAR FORAMEN** - posterior to carotid canal
  - **ANTERIOR** - internal petrosal sinus
  - **MIDDLE** - glossopharyngeal, vagus, and spinal accessory nerves.

- **POSTERIOR** - sigmoid sinus and meningeal branches from occipital and ascending pharyngeal arteries.
- **FORAMEN MAGNUM** - posterior to the basilar portion of the occipital lobe.
  - Medulla, spinal accessory nerves, vertebral arteries, ant. and post. spinal arteries
  - Bounded posteriorly by the **Occipital Condyles**.
- **CONDYLOID FOSSA** - perforated on one or both slides by the posterior condyloid canal; transmit emissary vein from the transverse sinus.
- **ANTERIOR CONDYLAR CANAL (HYPOGLOSSAL CANAL)** - transmits hypoglossal nerve and a meningeal artery.

INFERIOR OF THE SKULL - CALVARIA

- Concave
- With depressions for the convolutions of the cerebrum
- Furrows for the branches of the meningeal vessels
- **SUPERIOR SAGITTAL SINUS** - longitudinal groove, narrow anteriorly and posteriorly wide along the midline.

INFERIOR OF THE SKULL - FLOOR

- Divided into anterior, middle, and posterior

| FORAMENS                    | STRUCTURES                                                                                                     |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|
| CRIBRIFORM PLATE OF ETHMOID | Olfactory nerves                                                                                               |
| OPTIC FORAMEN               | Optic nerve, ophthalmic artery, meninges                                                                       |
| SUPERIOR ORBITAL FISSURE    | Oculomotor, trochlear, and abducens nerves; ophthalmic divisions of trigeminal nerve; superior ophthalmic vein |
| FORAMEN ROTUNDUM            | Maxillary division of trigeminal nerve, small artery, and vein                                                 |
| FORAMEN OVALE               | Mandibular division of trigeminal nerve and vein                                                               |
| FORAMEN LACERUM             | ICA and sympathetic plexus                                                                                     |
| FORAMEN SPINOSUM            | Middle meningeal artery and vein                                                                               |
| INTERNAL ACOUSTIC MEATUS    | Facial and vestibulocochlear nerves; internal auditory artery                                                  |
| JUGULAR FORAMEN             | Glossopharyngeal, vagus, and spinal accessory nerves; sigmoid sinus                                            |
| HYPOGLOSSAL CANAL           | Hypoglossal nerve                                                                                              |
| FORAMEN MAGNUM              | Medulla and meninges, spinal accessory nerve, vertebral arteries, anterior and posterior spinal arteries       |